Pivoted normalization results

Pivoted normalization at the slope frozen on Cranfield (s = 0.8) against cosine on the 12 EPC queries: MAP 0.711 → 0.729 (p = 0.13), with the same relevant articles in every top 10.

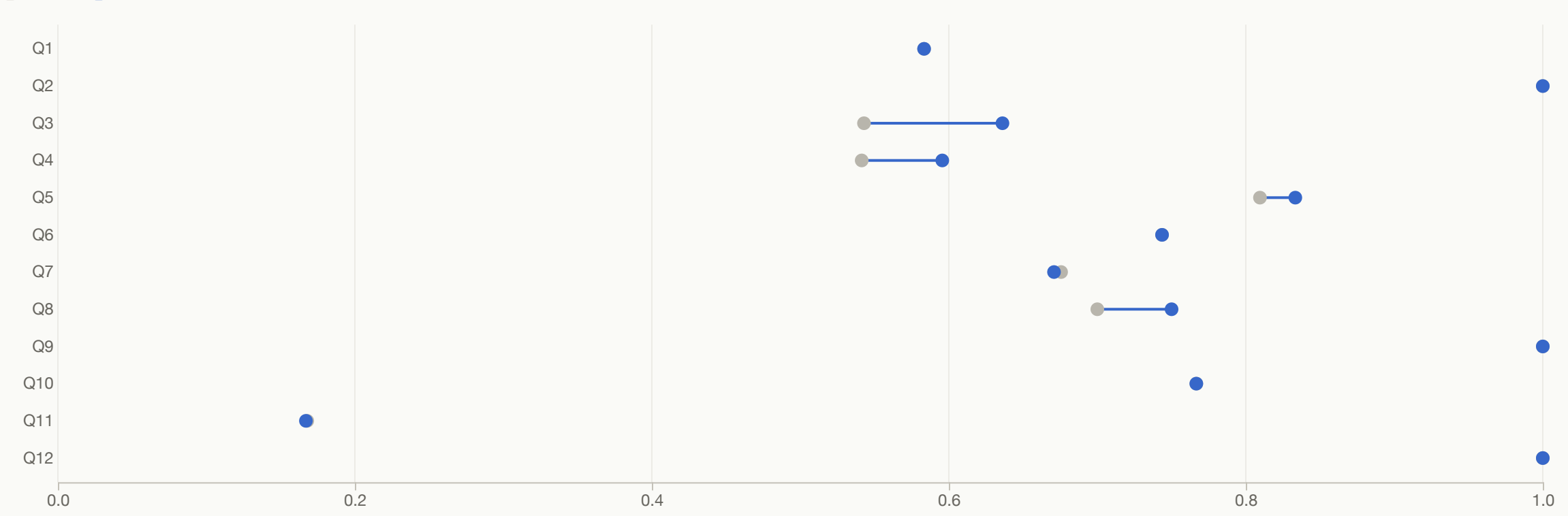
EPC retrieval quality (steps 35–37)

Each metric is averaged over the queries; AP uses each query's full ranking. Labels are the frozen `qrels_final.csv`.

MAP, cosine	MAP, pivoted	P@10, cosine / pivoted	R@10, cosine / pivoted	Queries where pivoted AP is	Randomization test, AP
0.711	0.729 +0.018 vs cosine	0.267 / 0.267	0.854 / 0.854	4 better 2 worse, 6 equal	p = 0.13 paired sign flips over queries

Average precision per query

Each line joins a query's cosine AP to its pivoted AP. Most queries do not move.



Per-query results

P@10 is capped by the number of relevant articles: a query with 2 relevant can reach at most 0.2.

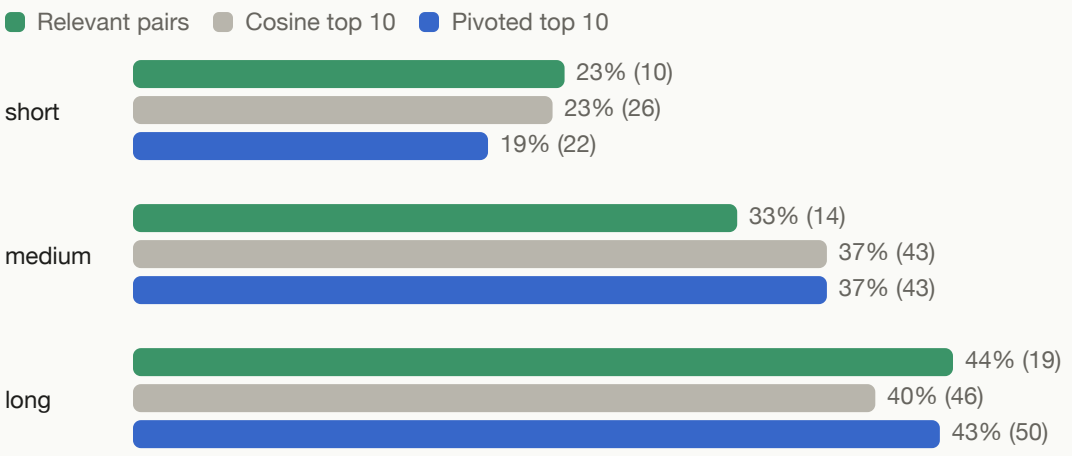
Query	Text	Relevant	P@10 cos	P@10 piv	R@10 cos	R@10 piv	AP cos	AP piv	ΔAP
Q1	placements	2	0.20	0.20	1.00	1.00	0.583	0.583	+0.000
Q2	HuEl	4	0.40	0.40	1.00	1.00	1.000	1.000	+0.000
Q3	fest budget	4	0.40	0.40	1.00	1.00	0.543	0.636	+0.093
Q4	online semester	6	0.50	0.50	0.83	0.83	0.541	0.596	+0.054
Q5	students union election process	3	0.30	0.30	1.00	1.00	0.810	0.833	+0.024
Q6	mental health counselling support	3	0.20	0.20	0.67	0.67	0.744	0.744	+0.000
Q7	packing list for freshers	4	0.30	0.30	0.75	0.75	0.676	0.671	−0.005
Q8	dual degree minor options	2	0.20	0.20	1.00	1.00	0.700	0.750	+0.050
Q9	what new sports facilities and tournaments did the sports secretary promise to students	2	0.20	0.20	1.00	1.00	1.000	1.000	+0.000
Q10	mess food quality complaints and what mess representatives promise to fix	3	0.30	0.30	1.00	1.00	0.767	0.767	+0.000
Q11	has the student union kept the promises from its election manifesto and what did it achieve	8	0.00	0.00	0.00	0.00	0.168	0.167	−0.001
Q12	why did students protest against the administration and how did the student union and administration respond	2	0.20	0.20	1.00	1.00	1.000	1.000	+0.000

Length groups (steps 38–39)

Articles are split into word-count tertiles (body words before stopword removal). Pivoting gives long articles 43% of top-10 slots, against 40% for cosine; long articles are 44% of relevant pairs.

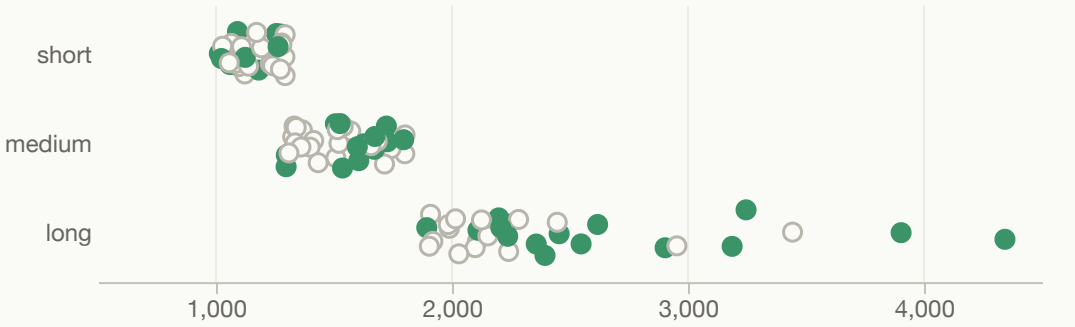
Share of top-10 slots and of relevant pairs

Bars are shares within each measure; the count follows in brackets.



Every article by length

Filled dots are relevant to at least one query.



Mean words, collection
1,709

Relevant
1,930

Cosine top 10
1,805

Pivoted top 10
1,841

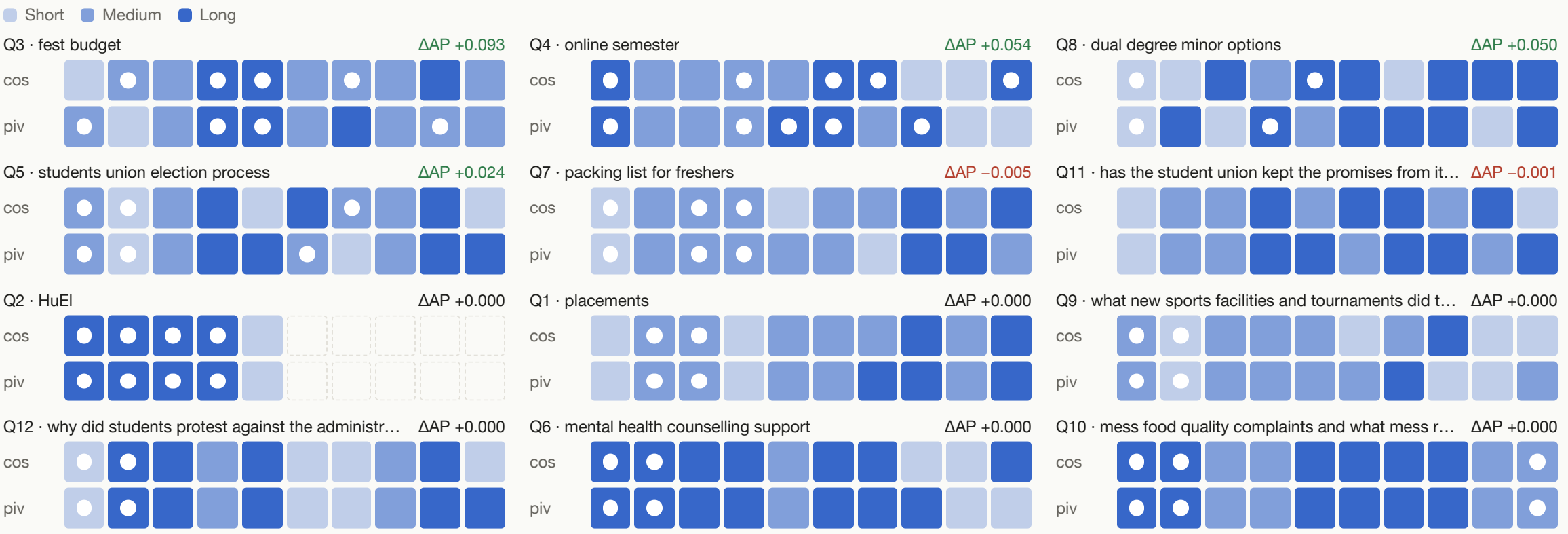
Groups

Top-10 slots total less than 120 because one query retrieves fewer than 10 articles.

Group	Articles	Min words	Max words	Relevant pairs	Top-10 cos	Top-10 piv	Relevant found cos	Relevant found piv
short	33	1,014	1,293	10	26	22	5	5
medium	34	1,297	1,801	14	43	43	11	11
long	33	1,893	4,343	19	46	50	16	16

Where the methods disagree (step 40)

Each square is a top-10 result, shaded by its length group; a dot marks a relevant article. Queries are ordered by how much AP changed.



Articles swapped into and out of the top 10

Every swap brings in a longer article; none changes which relevant articles are in the top 10.

Query	Entered	Words	Group	Relevant	Left	Words	Group	Relevant
Q1	7557	2,454	long	no	10685	1,571	medium	no
Q5	8341	1,904	long	no	8016	1,181	short	no
Q9	9930	1,722	medium	no	9136	1,014	short	no
Q11	8341	1,904	long	no	7282	1,124	short	no
Q12	9179	2,198	long	no	7452	1,108	short	no

Choosing s on Cranfield (steps 19–24)

The slope was chosen on Cranfield only, by MAP over all queries with at least one relevant document (grade ≥ 2), and checked with 5-fold cross-validation. EPC was not used.

Frozen slope s
0.8

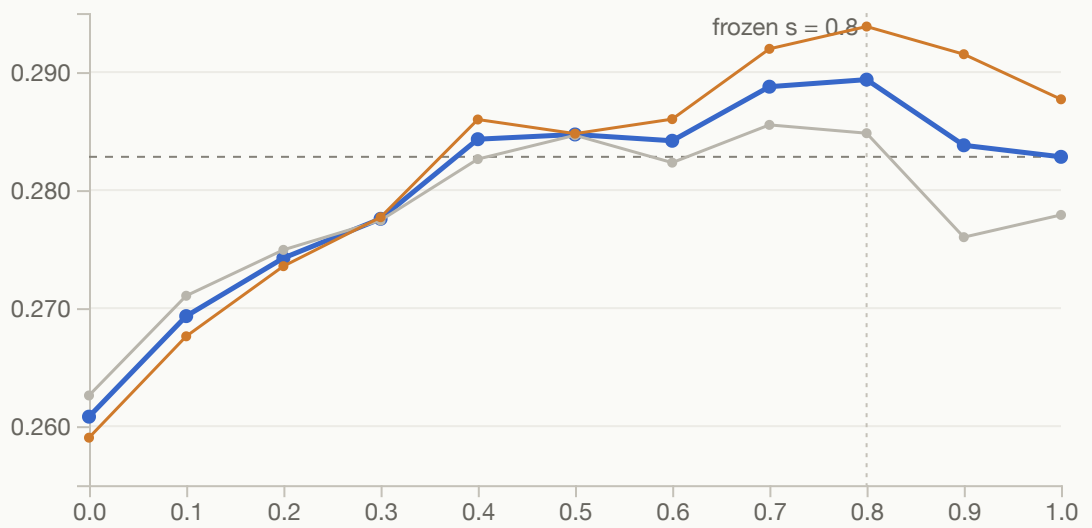
MAP at s, all queries
0.289
cosine 0.283

Out-of-fold MAP, pivoted
0.281
below cosine on unseen queries

MAP by slope

s = 1 is cosine. The dashed line is cosine over all queries.

All queries · Dev half · Held-out half · Cosine, all queries



Cross-validation folds

Each fold is scored with the slope chosen on the other four.

Fold	Chosen s	Queries	MAP cos	MAP piv
1	0.4	44	0.284	0.251
2	0.7	42	0.276	0.271
3	0.8	41	0.329	0.345
4	0.8	44	0.288	0.295
5	0.8	44	0.240	0.247

Engine versions

The engine moved to the paper's lnc.ltc weights and Porter stemming after the first freeze; s was refrozen.

Engine	Cosine	Best pivoted	s	Out-of-fold
TF-IDF both sides, no stemming	0.252	0.259	0.8	0.256
lnc.ltc + Porter stemming	0.283	0.289	0.8	0.281

Labels (steps 27–34)

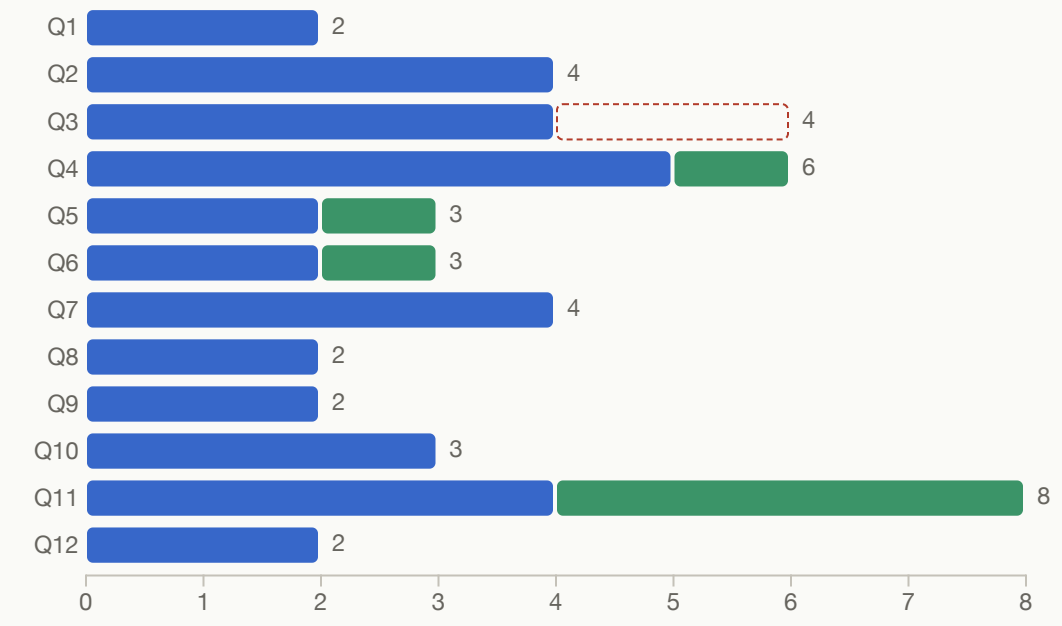
An LLM labelled every query–article pair. A person reviewed every pair the LLM flagged as uncertain, then the labels were frozen. The rest are unreviewed LLM labels.

Pairs labelled	Reviewed by hand	Changed by the reviewer	Relevant pairs, LLM → final	LLM positives not reviewed
1,200	36	9	38 → 43	29

Relevant articles per query

Kept: relevant to both the LLM and the final labels. Added and removed: changed by the reviewer.

Kept · Added by reviewer · Removed by reviewer



Labels the reviewer changed

Query	Article	LLM	Final
Q3	7784	relevant	not relevant
Q3	10490	relevant	not relevant
Q4	6977	not relevant	relevant
Q5	9930	not relevant	relevant
Q6	5170	not relevant	relevant
Q11	7429	not relevant	relevant
Q11	7434	not relevant	relevant
Q11	7988	not relevant	relevant
Q11	8016	not relevant	relevant

What was done

1. **Cranfield (19–24).** Loaded through ir_datasets; grades 2–4 count as relevant. A first single dev/held-out split failed its held-out check, so the slope is chosen over all queries and checked by 5-fold cross-validation. The slope was frozen in `frozen_slope.json`.
2. **Engine revision.** Documents now carry no IDF (the paper's lnc.ltc) and every word is Porter-stemmed. The slope was chosen again and stayed the same; the cross-validation check now fails narrowly, which the report must state.
3. **Queries and labels (25–29).** Queries with descriptions, a written relevance guideline and LLM labels for every pair, with reasons, excerpts and an uncertain flag. The LLM judged all queries for an article in one pass rather than one call per pair.
4. **Review and freeze (30–34).** Every uncertain pair was reviewed by hand in a review page that showed no rankings. The labels were frozen in `qrels_final.csv` with a checksum before this evaluation. The remaining LLM positives and a sample of negatives were not reviewed.
5. **Evaluation (35–40).** `run_epc.py` ranks every article for every query with both methods, computes the metrics, splits the collection into length tertiles and lists where the methods disagree. Details are in `docs/epc_evaluation.md`.
6. **Not done yet.** The attack experiments (41–51), the write-up and the optional extensions.

Exported from the Pivoted normalization results dashboard (claude.ai/artifact/SJDuvbf81pbZX6dA6JtZ5C), 24 October 2026. Data: run_epc.py and run_cranfield.py outputs on the frozen labels qrels_final.csv; engine with lnc.ltc weights and Porter stemming; s = 0.8.